

a) $u = x^3 - 3xy^2$

A function $u(x, y)$ is harmonic if it satisfies Laplace's equation.

Laplace's equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

First derivatives,

Given,

$u = x^3 - 3xy^2$ ——— (i)

Differentiating (i) with respect to x ,

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial}{\partial x} [x^3 - 3xy^2] \\ &= 3x^2 - 3y^2 \text{ ——— (ii)} \end{aligned}$$

Again differentiating (ii) with respect to x ,

$$\begin{aligned} \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) &= \frac{\partial}{\partial x} (3x^2 - 3y^2) \\ &= 6x \text{ ——— (iii)} \end{aligned}$$

Now differentiating (i) with respect to y ,

$$\begin{aligned} \frac{\partial u}{\partial y} &= \frac{\partial}{\partial y} [x^3 - 3xy^2] \\ &= -6xy \text{ ——— (iv)} \end{aligned}$$

Again differentiating (iv) with respect to y ,

$$\begin{aligned} \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) &= \frac{\partial}{\partial y} (-6xy) \\ &= -6x \text{ ——— (v)} \end{aligned}$$

Adding (ii) and (v),

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x - 6x = 0$$

Since ~~u(x,y) =~~ $u = x^3 - 3xy^2$ satisfies Laplace's equation

hence ~~u(x,y) =~~ $u = x^3 - 3xy^2$ is a harmonic function (Ans.)

b) $u = e^{-y} \sin x$ — (i)

Laplace's equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

First derivatives: with respect to x ,

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (e^{-y} \sin x)$$

$$= e^{-y} \frac{\partial}{\partial x} (\sin x) + \sin x \frac{\partial}{\partial x} (e^{-y})$$

$$= e^{-y} \cos x \quad \text{--- (ii)}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} (e^{-y} \cos x)$$

$$= e^{-y} \frac{\partial}{\partial x} \cos x + \cos x \frac{\partial}{\partial x} e^{-y}$$

$$= -e^{-y} \sin x \quad \text{--- (iii)}$$

~~Second derivative~~
with respect to y ,

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (e^{-y} \sin x)$$

$$= e^{-y} \frac{\partial}{\partial y} \sin x + \sin x \frac{\partial}{\partial y} e^{-y}$$

$$= -e^{-y} \sin x \quad \text{--- (iv)}$$

again,

$$\begin{aligned}
 \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} (-e^{-y} \sin x) \\
 &= -\left\{ e^{-y} \frac{\partial}{\partial y} \sin x + \sin x \frac{\partial}{\partial y} e^{-y} \right\} \\
 &= -\{-e^{-y} \sin x\} \\
 &= e^{-y} \sin x \quad \text{--- (v)}
 \end{aligned}$$

Adding (iii) and (v)

$$\begin{aligned}
 \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= -e^{-y} \sin x + e^{-y} \sin x \\
 &= 0 \\
 &\quad \text{(harmonic) (proved)}
 \end{aligned}$$

$$\text{Cf } u = e^x \cos y \quad \text{--- (i)}$$

with respect to x ,

$$\begin{aligned}
 \frac{\partial u}{\partial x} &= \frac{\partial}{\partial x} (e^x \cos y) \\
 &= e^x \frac{\partial}{\partial x} \cos y + \cos y \frac{\partial}{\partial x} e^x \\
 &= e^x \cos y \quad \text{--- (ii)}
 \end{aligned}$$

again,

$$\begin{aligned}
 \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} (e^x \cos y) \\
 &= e^x \frac{\partial}{\partial x} \cos y + \cos y \frac{\partial}{\partial x} e^x \\
 &= e^x \cos y \quad \text{--- (iii)}
 \end{aligned}$$

with respect to y ,

$$\begin{aligned}\frac{\delta u}{\delta y} &= \frac{\delta}{\delta y} (e^x \cos y) \\ &= e^x \frac{\delta}{\delta y} \cos y + \cos y \frac{\delta}{\delta y} e^x \\ &= 0 - e^x \sin y \quad \text{--- (iv)}\end{aligned}$$

again,

$$\begin{aligned}\frac{\delta^2 u}{\delta y^2} &= \frac{\delta}{\delta y} (-e^x \sin y) \\ &= -\left\{ e^x \frac{\delta}{\delta y} \sin y + \sin y \frac{\delta}{\delta y} e^x \right\} \\ &= -\{ e^x \cos y \} \\ &= -e^x \cos y \quad \text{--- (v)}\end{aligned}$$

adding (iii) and (v) $\Rightarrow$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = e^x \cos y - e^x \cos y = 0$$

(proved harmonic)

d) $u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2$ --- (i)

with respect to x ,

$$\begin{aligned}\frac{\delta u}{\delta x} &= \frac{\delta}{\delta x} (3x^2y + 2x^2 - y^3 - 2y^2) \\ &= 6xy + 4x \quad \text{--- (ii)}\end{aligned}$$

again,

$$\begin{aligned}\frac{\delta^2 u}{\delta x^2} &= \frac{\delta}{\delta x} (6xy + 4x) \\ &= 6y + 4 \quad \text{--- (iii)}\end{aligned}$$

with respect to y

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (3x^2y + 2x^2 - y^3 - 2y^2)$$

$$= 3x^2 - 3y^2 - 4y \quad \text{--- (iv)}$$

again,

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y} (3x^2 - 3y^2 - 4y)$$

$$= -6y - 4$$

$$= -(6y + 4) \quad \text{--- (v)}$$

adding (iii) and (v) $\Rightarrow$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = (6y + 4) - (6y + 4)$$

$$= 0$$

(harmonic)

e] $u(x, y) = e^x (x \cos y - y \sin y) \quad \text{--- (i)}$

with respect to x ,

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \{ e^x (x \cos y - y \sin y) \}$$

$$= e^x \frac{\partial}{\partial x} (x \cos y - y \sin y) + (x \cos y - y \sin y) \frac{\partial}{\partial x} e^x$$

$$= e^x \left[\frac{\partial}{\partial x} (x \cos y) - \frac{\partial}{\partial x} (y \sin y) \right] + (x \cos y - y \sin y) e^x$$

$$= e^x \left[\{ x \frac{\partial}{\partial x} \cos y + \cos y \frac{\partial}{\partial x} x \} + \{ y \frac{\partial}{\partial x} \sin y + \sin y \frac{\partial}{\partial x} y \} \right]$$

$$+ (x \cos y - y \sin y) e^x$$

$$= e^x [\cos y] + (x \cos y - y \sin y) e^x$$

$$= e^x (\cos y + x \cos y - y \sin y) \quad \text{--- (ii)}$$

again,

Nasif

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} \left\{ e^x (x \cos y + y \cos x - x \sin y) \right\} \\ &= e^x \frac{\partial}{\partial x} (x \cos y + y \cos x - x \sin y) + (x \cos y + y \cos x - x \sin y) e^x \\ &= e^x \left\{ 0 + (x \frac{\partial}{\partial x} \cos y + \cos y \frac{\partial}{\partial x} x) - (y \frac{\partial}{\partial x} \sin y + \sin y \frac{\partial}{\partial x} x) \right\} \\ &\quad + (x \cos y + y \cos x - x \sin y) e^x \\ &= e^x \left\{ (0 + \cos y) - 0 \right\} + (x \cos y + y \cos x - x \sin y) e^x \\ &= e^x \cos y + (x \cos y + y \cos x - x \sin y) e^x \\ &= e^x (x \cos y - y \sin y + 2 \cos y) \quad \text{--- (iii)} \end{aligned}$$

with respect to y,

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (x \cos y + y \cos x - x \sin y)$$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \left\{ e^x (x \cos y - y \sin y) \right\}$$

--- বাকি অংশ করতে হবে আলাদা করে,

$$\begin{aligned} &= x \cos y - y \sin y \\ &= e^x (-x \sin y - y \cos y - \sin y) \quad \text{--- (iv)} \end{aligned}$$

again,

$$\begin{aligned} \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} \left\{ e^x (-x \sin y - y \cos y - \sin y) \right\} \\ &= e^x (-x \cos y + y \sin y - 2 \cos y) \quad \text{--- (v)} \end{aligned}$$

adding (iii) and (v) $\Rightarrow$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = e^x (x \cos y - y \sin y + x \cos y) + e^x (-x \cos y + y \sin y - 2 \cos y)$$

$$= 0 \quad (\text{harmonic}).$$

$$u = y^3 - 3x^2 y \quad \text{--- (i)}$$

with respect to x ,

$$\frac{\delta u}{\delta x} = \frac{\delta}{\delta x} (y^3 - 3x^2 y)$$

$$= -6xy$$

$$\text{--- (ii)}$$

again,

$$\frac{\delta^2 u}{\delta x^2} = \frac{\delta}{\delta x} (-6xy)$$

$$= -6y \quad \text{--- (iii)}$$

with respect to y ,

$$\frac{\delta u}{\delta y} = \frac{\delta}{\delta y} (y^3 - 3x^2 y)$$

$$= 3y^2 - 3x^2 \quad \text{--- (iv)}$$

again,

$$\frac{\delta^2 u}{\delta y^2} = \frac{\delta}{\delta y} (3y^2 - 3x^2)$$

$$= 6y \quad \text{--- (v)}$$

adding (iii) and (v)

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = -6y + 6y$$

$$= 0$$

(harmonic)

7] $u = 2x(1-y)$ ——— (i)

with respect to x ,

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial}{\partial x} \{2x(1-y)\} \\ &= \frac{\partial}{\partial x} (2x - 2xy) \\ &= 2 - 2y \text{ ——— (ii)}\end{aligned}$$

again,

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} (2 - 2y) \\ &= 0 - 0 \\ &= 0 \text{ ——— (iii)}\end{aligned}$$

with respect to y ,

$$\begin{aligned}\frac{\partial u}{\partial y} &= \frac{\partial}{\partial y} \{2x(1-y)\} \\ &= \frac{\partial}{\partial y} (2x - 2xy) \\ &= -2x \text{ ——— (iv)}\end{aligned}$$

again,

$$\begin{aligned}\frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} (-2x) \\ &= 0 \text{ ——— (v)}\end{aligned}$$

adding (iii) and (v)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 + 0$$

$= 0$
(harmonic)
(satisfies)

Narif

h) $u(x, y) = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ ——— (i) ——— Nasif

with respect to x ,

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial}{\partial x} (x^3 - 3xy^2 + 3x^2 - 3y^2 + 1) \\ &= 3x^2 - 3y^2 + 6x - 0 + 0 \\ &= 3x^2 - 3y^2 + 6x \text{ ——— (ii) ———}\end{aligned}$$

again,

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} (3x^2 - 3y^2 + 6x) \\ &= 6x - 0 + 6 \\ &= 6x + 6 \text{ ——— (iii) ———}\end{aligned}$$

with respect to y ,

$$\begin{aligned}\frac{\partial u}{\partial y} &= \frac{\partial}{\partial y} (x^3 - 3xy^2 + 3x^2 - 3y^2 + 1) \\ &= 0 - 6xy + 0 - 6y + 0 \\ &= -6xy - 6y \text{ ——— (iv) ———}\end{aligned}$$

again,

$$\begin{aligned}\frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} (-6xy - 6y) \\ &= -6x - 6 \text{ ——— (v) ———}\end{aligned}$$

adding (iii) and (v) $\Rightarrow$

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= 6x + 6 - 6x - 6 \\ &= 0 \\ &\text{(harmonic)}\end{aligned}$$

$$\text{i)} \quad u(x, y) = x^2 - y^2 - 2xy - 2x + 3y \quad \text{--- (i)}$$

with respect to x ,

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (x^2 - y^2 - 2xy - 2x + 3y)$$

$$= 2x - 0 - 2y - 2 + 0$$

$$= 2x - 2y - 2 \quad \text{--- (ii)}$$

again,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} (2x - 2y - 2)$$

$$= 2 - 0 - 0$$

$$= 2 \quad \text{--- (iii)}$$

with respect to y ,

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (x^2 - y^2 - 2xy - 2x + 3y)$$

$$= 0 - 2y - 2x + 0 + 3$$

$$= -2y - 2x + 3 \quad \text{--- (iv)}$$

again,

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y} (-2y - 2x + 3)$$

$$= -2 + 0 + 0$$

$$= -2 \quad \text{--- (v)}$$

adding (iii) and (v) $\Rightarrow$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 - 2$$

$$= 0$$

(harmonic)